

Тензор инерции

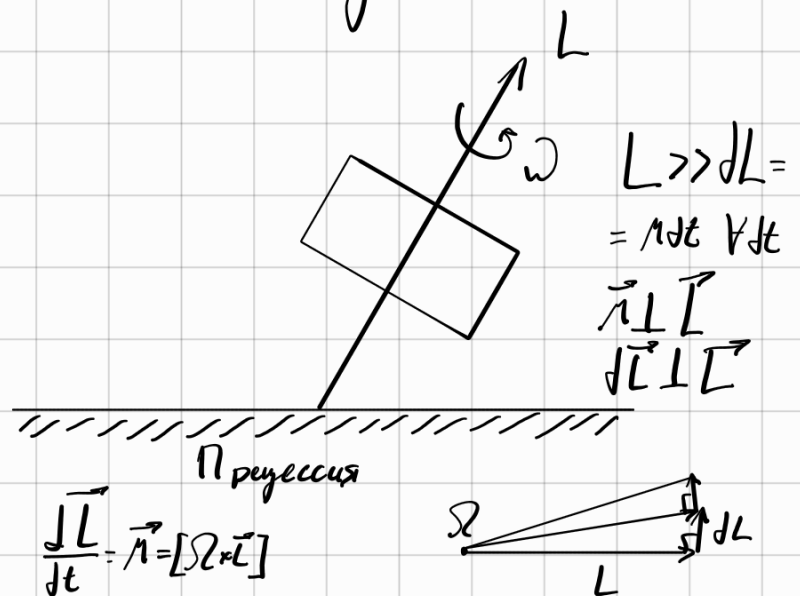
$$I_{xy} = \int (y^2 + z^2) dm$$

$$I_{xy} = - \int x y \, dm = I_{yx}$$

TEMP - curr ramp. $\Rightarrow \exists$ value on:

$$\begin{pmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{pmatrix}$$

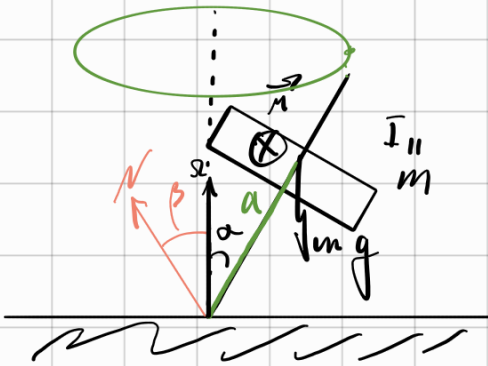
Эмпсон и Керуи



11.1

$$N_y = mg$$

$$N = mg a \sin \alpha$$



$$N_x = m \Omega^2 a \sin \alpha$$

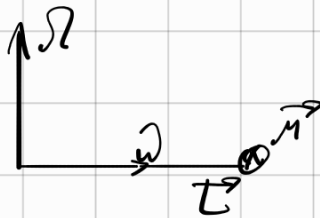
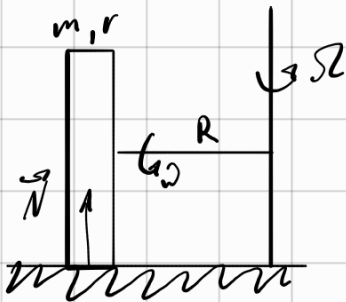
$$\tan \beta = \frac{N_x}{N_y} = \frac{\Omega^2 a \sin \alpha}{g}$$

$$L = I_{\parallel} \Omega$$

$$M = \Omega \cdot I_{\parallel} \cdot \sin \alpha$$

$$\Omega = \frac{mg a}{I_{\parallel}}$$

11.14



$$M = FR$$

$$\Omega \cdot L_c = \Omega \frac{mR^2}{2} \Omega$$

$$\Omega = \Omega \frac{R}{r}$$

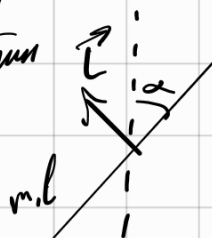
$$F = \frac{m \Omega^2 r}{2}$$

$$N = mg + \frac{m \Omega^2 r}{2}$$

11.24

$$n = 3000 \frac{1}{\text{mm}}$$

$$\alpha = 89.9^\circ$$



$$dm = \frac{m}{l} dx$$

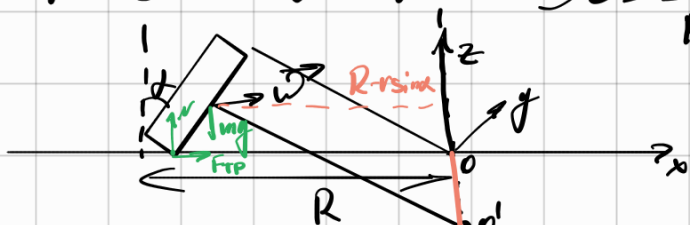
$$dL = x dp = x dm \Omega \sin \alpha$$

$$L = \int_{-\frac{l}{2}}^{\frac{l}{2}} \frac{m}{l} \Omega \sin \alpha x^2 dx = \frac{m}{l} \Omega \sin \alpha \frac{x^3}{3} \Big|_{-\frac{l}{2}}^{\frac{l}{2}} =$$

$$= \frac{m l^2}{12} \Omega^2 \sin \alpha$$

$$M = \Omega L \cdot \sin\left(\frac{\pi}{2} - \alpha\right) = \frac{m l^2}{12} \Omega^2 \sin \alpha \sin\left(\frac{\pi}{2} - \alpha\right)$$

11. 18 $\omega = \frac{v}{r}$ $\Omega = \frac{v}{R}$



$$\vec{N} = [\vec{\Omega} \times \vec{L}] = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 0 & -\Omega \\ -L_y & 0 & L_z \end{vmatrix} = \vec{j} \Omega L_x$$

$$m \Omega^2 (R - r \sin \alpha) = F_{TP}$$

$$N = mg$$

$$M_N = mg R$$

$$M_{mg} = -mg (R - r \sin \alpha)$$

$$M_{TP} = m \Omega^2 (R - r \sin \alpha) r_{od} = m \Omega^2 (R - r \sin \alpha) \frac{R \sin \alpha - r}{\cos \alpha}$$

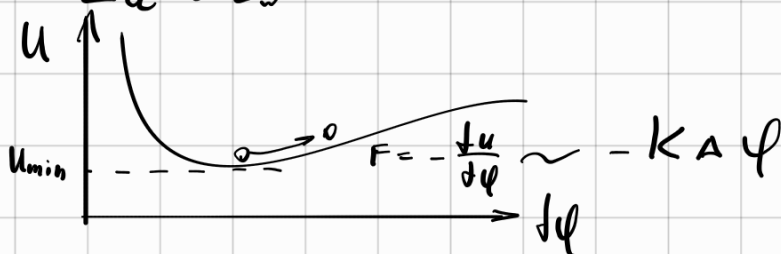
$$\Omega L_x = - \frac{(R - r \sin \alpha)(R \sin \alpha - r)}{\cos \alpha} m \Omega^2 - (R - r \sin \alpha) mg + mg R$$

$$\vec{L} = \vec{L}_{||} + \vec{L}_{\perp}$$

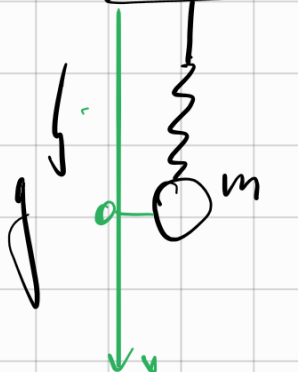
$$L_{||} = I_{||} \omega_{||} = m r^2 (\omega + \Omega_{||})$$

$$L_{\perp} = I_{\perp} \omega_{\perp} = \left(\frac{m r^2}{2} + m \left(\frac{(R - r \sin \alpha)^2}{\cos \alpha} \right) \right) \Omega_{\perp}$$

$$L_{\Omega} \sim L \omega$$



543



$$F = K(\ell - \ell_0)^2$$

$$mg = K \Delta \ell_0^2$$

$$\Delta \ell_0 = \sqrt{\frac{mg}{K}}$$

$$\vec{F}_y = K(\Delta \ell_0 + x)^2$$

$$m \ddot{x} = mg - K(\Delta \ell_0 + x)^2$$

$$m\ddot{x} = -2K\Delta l_0 x$$

$$\ddot{x} + \frac{2K\Delta l_0}{m} x = 0$$

$$\omega = \sqrt{\frac{2K\Delta l_0}{m}} = \sqrt{\frac{4K\ell}{m}}$$

$$U = \frac{A\dot{\varphi}^2}{2}$$

$$K = B \frac{\dot{\varphi}^2}{2}$$

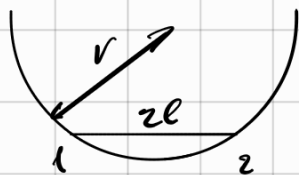
$$E = A \frac{\dot{\varphi}^2}{2} + B \frac{\dot{\varphi}^2}{2}$$

$$E = 0 = A\varphi\dot{\varphi} + B\dot{\varphi}\ddot{\varphi}$$

$$0 = A\varphi + B\ddot{\varphi}$$

$$\ddot{\varphi} + \frac{A}{B}\varphi = 0 \quad \omega = \sqrt{\frac{A}{B}} \quad T = 2\pi\sqrt{\frac{B}{A}}$$

5.22



1) $\perp \rho_{uc}$



$$\rho = \sqrt{r^2 - \ell^2}$$

$$U = 2mg(\rho - \rho \cos \varphi) \approx 2mg\rho \frac{\varphi^2}{2}$$

$$K = \frac{2m\rho^2}{2} \dot{\varphi}^2$$

$$T = 2\pi\sqrt{\frac{B}{A}} = 2\pi\sqrt{\frac{2m\rho^2}{2mg\rho}} = 2\pi\sqrt{\frac{r^2 - \ell^2}{g}}$$

2)



$$U = 2mg\rho \frac{\varphi^2}{2}$$

$$K = 2m r^2 \frac{\varphi^2}{2}$$

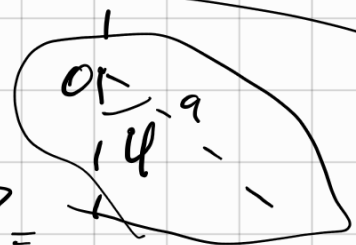
$$T = 2\pi\sqrt{\frac{K}{U}}$$

Задача 5.23. МАСТНИК

$$I\ddot{\varphi} = -mga \sin \varphi$$

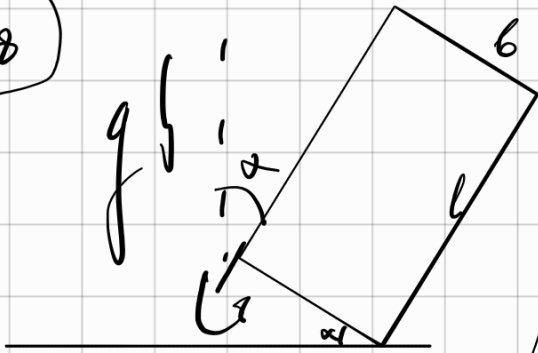
$$\ddot{\varphi} + \frac{mga}{I}\varphi = 0$$

$$T = 2\pi\sqrt{\frac{I}{mga}}$$



$$= 2\pi \sqrt{\frac{L_{OP}}{g}} \quad L_{OP} = \frac{I}{ma}$$

1048



T-?

$$\frac{b}{2} (1 - \cos \varphi)$$

$$\frac{b}{2} \sin \alpha (1 - \cos \varphi)$$

$$U \approx mg \frac{b}{2} \sin \alpha \frac{\varphi^2}{2}$$

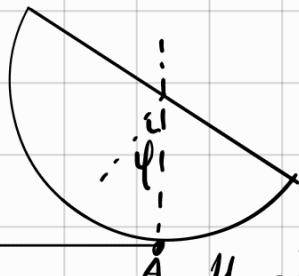
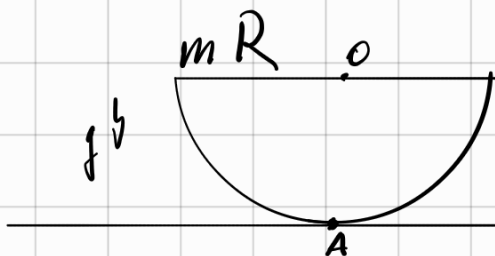
$$K = I \frac{\dot{\varphi}^2}{2}; \quad I = \frac{mb^3}{3}$$

$$T = 2\pi \sqrt{\frac{2b}{3g \sin \alpha}}$$

$$E = \frac{A\dot{\varphi}}{2} + \frac{B\dot{\psi}}{2}$$

$$T = 2\pi \sqrt{\frac{B}{A}}$$

10.84



$$K = I_A \frac{\dot{\varphi}^2}{2} \quad I_A = I_C + m(R - y_c)^2 + (y_c)^2$$

$$I_O = I_C + m y_c^2 = \frac{m R^2}{2}$$

$$I_A = \frac{m R^2}{2} - m y_c^2 + m(R - y_c)^2$$

$$= \frac{3}{2} m R^2 - \frac{8}{3\pi} m R^2$$

$$T = 2\pi \sqrt{\frac{m R^2 (\frac{3}{2} - \frac{8}{3\pi})}{\frac{4}{3\pi} m g R}} = 2\pi \sqrt{\frac{R}{g} \frac{3\pi - 16}{8}}$$

$$U = m g y_c (1 - \cos \varphi) = m g y_c \frac{\varphi^2}{2}$$

$$y_c = \frac{1}{m} \int_0^R y dm \quad dm = \frac{2 \sqrt{R^2 - y^2} m dy}{\pi R^2}$$

$$y_c = \frac{2}{\pi R^2} \int_0^R y \sqrt{R^2 - y^2} dy = \frac{-2}{\pi R^2} \int_0^R x dx$$

$$R^2 - y^2 = x$$

$$dx = -2y dy$$

$$= -\frac{2}{\pi R^2} \frac{2}{3} x^{\frac{3}{2}} \Big|_0^R = \frac{4}{3\pi} R$$

$$\vec{a} = \frac{\vec{F}}{m} \quad \text{в УЧД}$$

$$m\vec{a} = \vec{F} + \vec{F}_{\text{ин}} \quad \vec{F}_{\text{ин}} = -m\vec{a}_{\text{учо}}$$

0 12.1

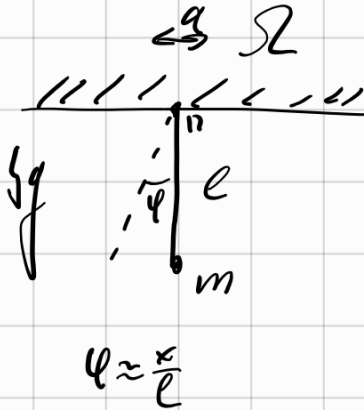
Дано:

$$l = 9,8 \text{ см}$$

$$\Omega = 11 \frac{\text{рад}}{\text{с}}$$

$$\alpha = 1 \text{ м}$$

Найти: A



$$x_n = a \cos \Omega t$$

$$F_{\text{ин}} = -m \ddot{x}_n = m a \Omega^2 \cos \Omega t$$

$$m \ddot{x} = m a \Omega^2 \cos \Omega t - T \cdot \frac{x}{l} \quad \ominus$$

$$\psi \text{ мал } \cos \psi \approx 1$$

$$\ominus m a \Omega^2 \cos \Omega t - mg \frac{x}{l}$$

$$\ddot{x} + g \frac{x}{l} = a \Omega^2 \cos \Omega t$$

неоднородное Д.У.

общее решение: Σ общее решение однородного

+ частное решение неоднородного
в виде $A \cos(\Omega t + \psi)$

⊕

$$-\Omega^2 A \cos(\Omega t + \psi) + \omega_0^2 A \cos(\Omega t + \psi) = a \Omega^2 \cos \Omega t$$

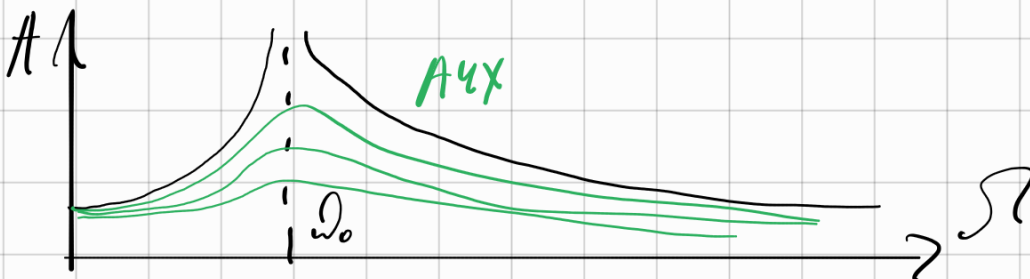
$$-\Omega^2 A \cos \Omega t \cos \psi + \Omega^2 A \sin \Omega t \sin \psi + \omega_0^2 A \cos \Omega t \cos \psi -$$

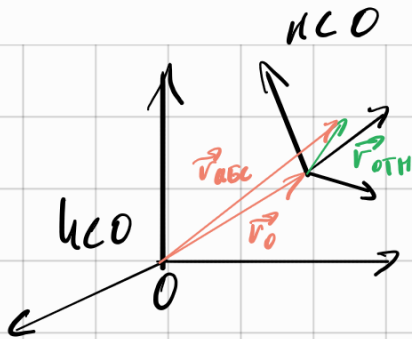
$$- \omega_0^2 A \sin \Omega t \sin \psi = a \Omega^2 \cos \Omega t$$

$$\left\{ \begin{array}{l} -\Omega^2 A \cos \psi + \omega_0^2 A \cos \psi = \Omega^2 a \\ \Omega^2 A \sin \psi - \omega_0^2 A \sin \psi = 0 \end{array} \right.$$

$$\sin \psi = 0 \quad \cos \psi = \pm 1$$

$$A = \frac{a \Omega^2}{\pm (\omega_0^2 - \Omega^2)} \quad \begin{array}{l} \cos \psi = -1 \\ A > 0 \end{array} = \frac{a}{1 - \frac{\Omega^2}{\omega_0^2}} = 5,8 \text{ мм}$$





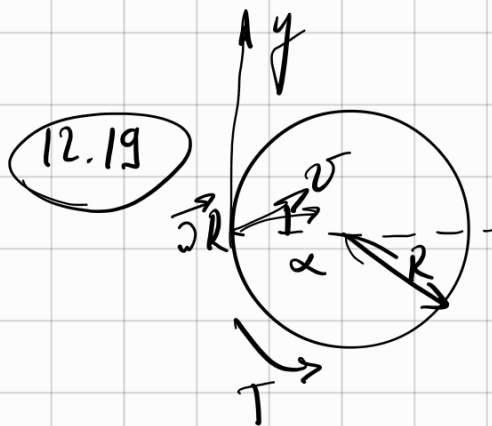
$$\vec{r}_{ABC} = \vec{r}_{OTM} + \vec{r}_O$$

$$\vec{v}_{ABC} = \vec{v}_{OTM} + [\vec{\omega} \times \vec{r}_{OTM}] + \vec{v}_O$$

$$\vec{a}_{ABC} = \vec{a}_{OTM} + \underbrace{\vec{a}_O}_{\vec{a}_{\text{пер}}} + [\vec{\omega} \times \vec{v}] + [\vec{\omega} \times [\vec{\omega} \times \vec{r}_{OTM}]] + 2[\vec{\omega} \times \vec{v}_{OTM}]$$

$$F_y \delta = m \omega^2 r_{OTM}$$

$$\vec{F}_{\text{кор}} = 2m[\vec{v}_{OTM} \times \vec{\omega}] = -m\vec{a}_{\text{кор}}$$



$$v \gg \omega R$$

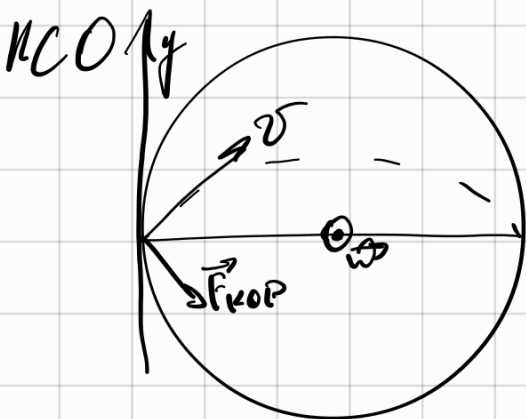
$$\alpha \ll 1$$

$$t \approx \frac{2R \cos \alpha}{|\vec{v} + \vec{\omega} R|} = \frac{2R(1 - \frac{\alpha^2}{2})}{v} \approx \frac{2R}{v}$$

$$|f_{\text{H}}| = (v \sin \alpha - \omega R) t \approx \omega R t$$

$$\sin \alpha \approx \alpha = \frac{2\omega R}{v} \quad T = \frac{2\pi}{\omega}$$

$$\alpha = \frac{4\pi R}{vT} \approx 0.0209 \text{ рад} = 1.2^\circ$$



$$F_{\text{кор}} = 2m\omega v$$

$$F_y \delta = m\omega^2 x \leq m\omega^2 R \ll m\omega v$$

$$m\ddot{y} = -2m\omega v$$

$$y(t) = v \sin \alpha - 2\omega v t$$

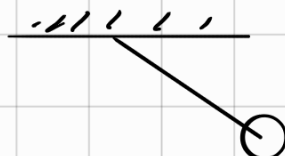
$$y(t) = v \sin \alpha t - 2\omega v t^2$$

$$t = \frac{2R}{v} \quad y(t) = 0$$

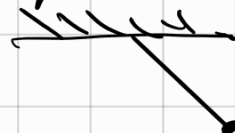
$$\alpha = \frac{\omega v}{v} \cdot \frac{2R}{v} = \frac{2\omega R}{v}$$

10.43

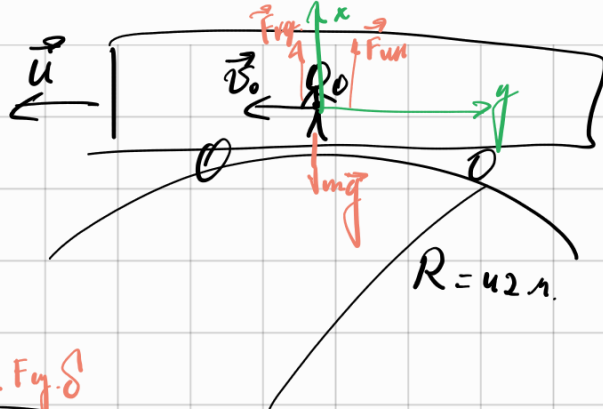
$$K_f = \frac{1}{2} \frac{v^2}{\omega^2} = \frac{1}{2} \frac{(\psi \epsilon)^2}{\omega^2}$$



Риск гравитации несут только



12.27



Ocu bany buyme c gburu.
abmodya $\omega = \frac{u}{R}$

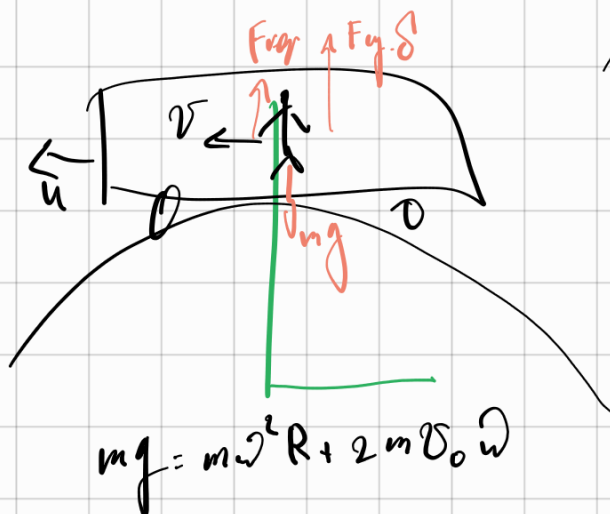
$$F_y \cdot \delta = 0$$

$$F_{\text{Cor}} = 2m [v \times \omega]$$

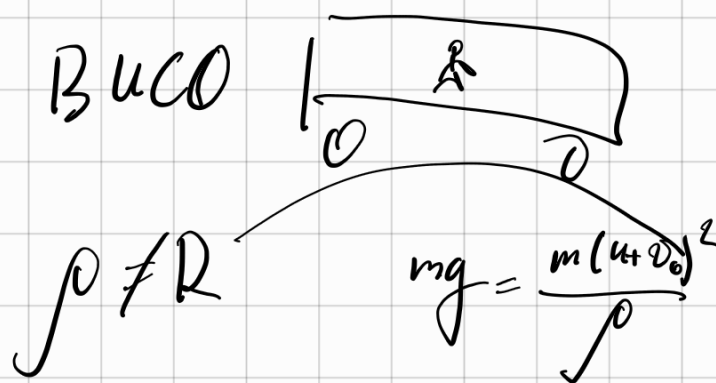
$$F_{\text{un}} = m \frac{u^2}{R}$$

$$mg = m \frac{u^2}{R} + 2m v_0 \cdot \frac{u}{R}$$

$$v_0 = \frac{gR - u^2}{2u} \approx 0.23 \text{ m/s}$$



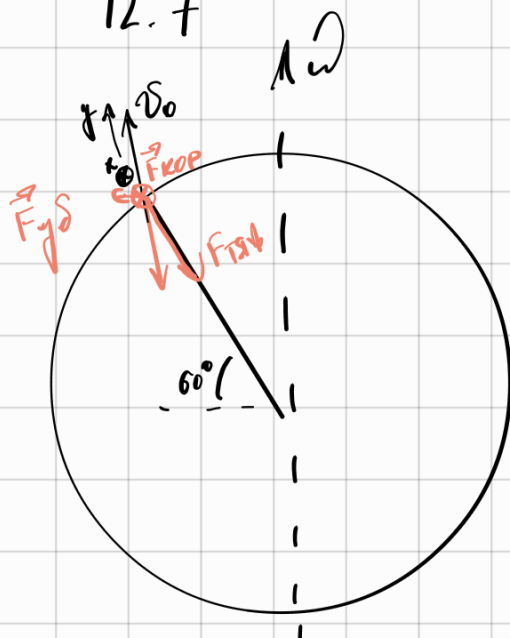
$$mg = m \omega^2 R + 2m v_0 \omega$$



$$\rho \neq R$$

$$mg = \frac{m(u + v_0)^2}{\rho}$$

12.7

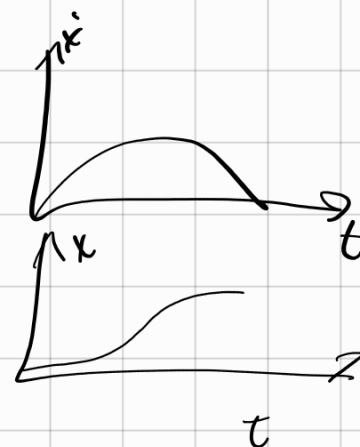


$$m \ddot{y} \approx mg$$

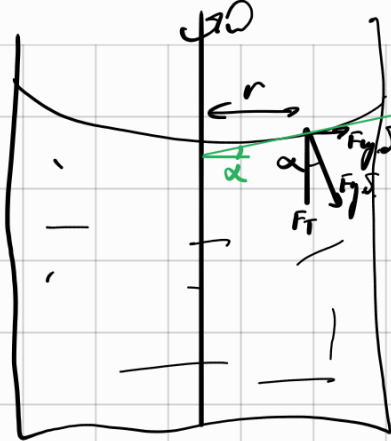
$$m \ddot{x} = 2m \dot{y} \omega \sin\left(\frac{\pi}{2} - \varphi\right)$$

$$\dot{y} = v_0 - g t$$

gbanuon



14.16



$$z(r) = ?$$

$$\tan \alpha = \frac{F_T S}{F_N} = \frac{\partial z}{\partial r}$$

$$z(r) = h + \frac{\omega^2 r^2}{2g}$$



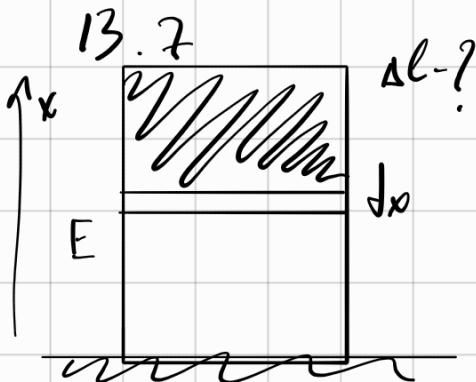
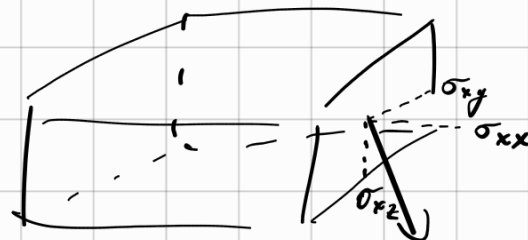
$$F = kx$$

$$\sigma = \frac{F}{S} \text{ механическое напряжение}$$

$$\epsilon = \frac{x}{l} \text{ относительная деформация}$$

$$\sigma = \epsilon \cdot E \text{ закон Гука}$$

$$\mu = -\frac{\epsilon_y}{\epsilon_x} \text{ коэфф. Пуассона}$$



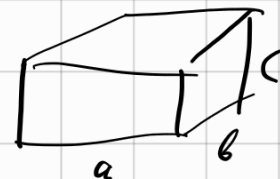
$$P(x) = P(1 - \frac{x}{h})$$

$$\frac{dE}{dx} = \frac{d(\Delta l)}{dx} = \frac{P(x)}{SE}$$

$$\Delta l = \int_0^h \frac{P}{SE} \left(1 - \frac{x}{h}\right) dx = \frac{Ph}{2SE}$$

$$\frac{\Delta V}{V} = \frac{\Delta a}{a} + \frac{\Delta b}{b} + \frac{\Delta c}{c} = \frac{\Delta a}{a} (1 - 2\mu)$$

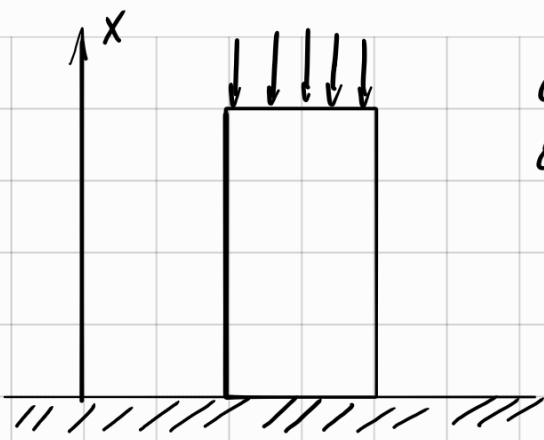
$$\frac{\Delta V}{V} = \frac{P}{2SE} (1 - 2\mu)$$



$$W = \frac{kx^2}{2}$$

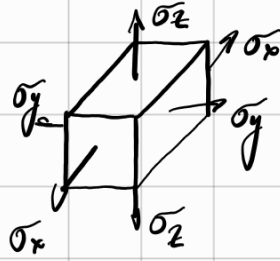
$$w = \frac{W}{V} = E \frac{\epsilon^2}{2} = \frac{\sigma^2}{2E} \text{ удельная потенциальная энергия}$$

$$W = \int_0^h w(x) S dx = \frac{P^2 h}{6SE}$$



$$\epsilon_x = \frac{\sigma_x}{E}$$

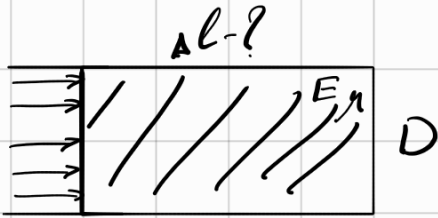
$$\epsilon_y = \epsilon_z = -\mu \epsilon_x = -\mu \frac{\sigma_x}{E}$$



$$\epsilon_x = \frac{\sigma_x}{E} - \mu \frac{\sigma_y}{E} - \mu \frac{\sigma_z}{E} = \frac{1}{E}(\sigma_x - \mu(\sigma_y + \sigma_z))$$

$$\epsilon_y = \frac{1}{E}(\sigma_y - \mu(\sigma_x + \sigma_z))$$

$$\epsilon_z = \frac{1}{E}(\sigma_z - \mu(\sigma_y + \sigma_x))$$



$$\sigma_x = \frac{4F}{\pi D^2}$$

$$\epsilon_x = \frac{4F}{E \pi D^2} - \frac{1}{E} 2\sigma$$

$$\epsilon_y = \epsilon_z = 0 = \frac{1}{E}(\sigma - \mu(\frac{4F}{\pi D^2} + \sigma))$$

$$\sigma_y = \sigma_z = 0$$

$$\frac{\sigma}{E} = \frac{1}{E} \frac{4F}{\pi D^2} + \frac{\sigma}{E} \mu$$

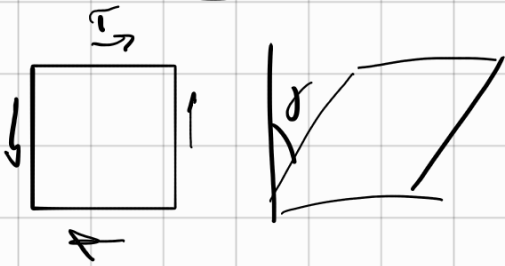
$$\sigma \left(\frac{1}{E} - \frac{\mu}{E} \right) = \frac{1}{E} \frac{4F}{\pi D^2}$$

$$\sigma = \frac{1}{\pi D^2 (1 - \mu)} \frac{4F}{\pi D^2}$$

$$\epsilon_x = \frac{4F}{\pi D^2 E} - \frac{2\mu}{E} \frac{4F}{\pi D^2 (1 - \mu)} =$$

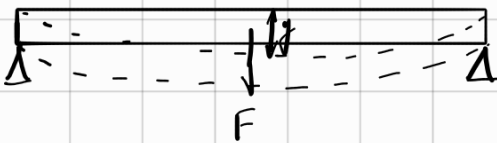
$$= \frac{4F}{\pi D^2 E} \frac{1 - \mu - 2\mu^2}{1 - \mu} = \frac{1}{E}$$

коэффициент сжатия

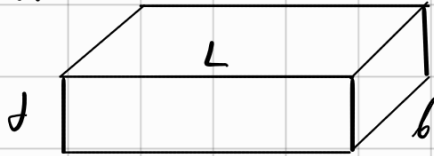


$$\tau = \sigma$$

коэффициент сжатия



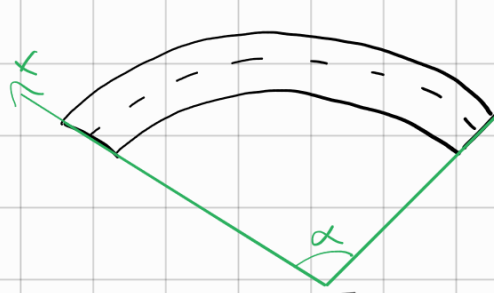
13.41



Канало: $\alpha = 2\pi$ $R = \frac{L}{2\pi}$

$$\int W(x) = \frac{2\pi^2 E b}{L} x^2 dx$$

$$W = \int_{-\frac{d}{2}}^{\frac{d}{2}} \frac{2\pi^2 E b}{L} x^2 dx = \frac{2\pi^2 E b}{L} \frac{x^3}{3} \Big|_{-\frac{d}{2}}^{\frac{d}{2}} = \frac{\pi^2 E b d^3}{6L}$$



$$L = R\alpha$$

$$L(x) = (R+x)\alpha$$

$$\epsilon(x) = \frac{L(x) - L}{L} = \frac{x}{R}$$

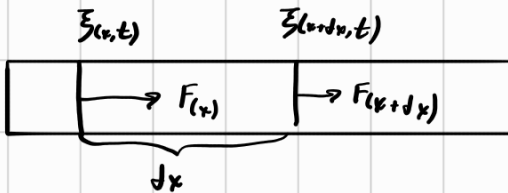
$$\sigma(x) = E \frac{x}{R}$$

Плотность энергии упругой деформации:

$$w(x) = \frac{\sigma(x)^2}{2E} = E \frac{x^2}{2R^2}$$

$$\int W(x) = w(x) \cdot L \cdot b \cdot dx = E L b \frac{x^2}{2R^2} dx$$

Волны



$$F(x+dx) \approx F(x) + \frac{\partial F}{\partial x} dx$$

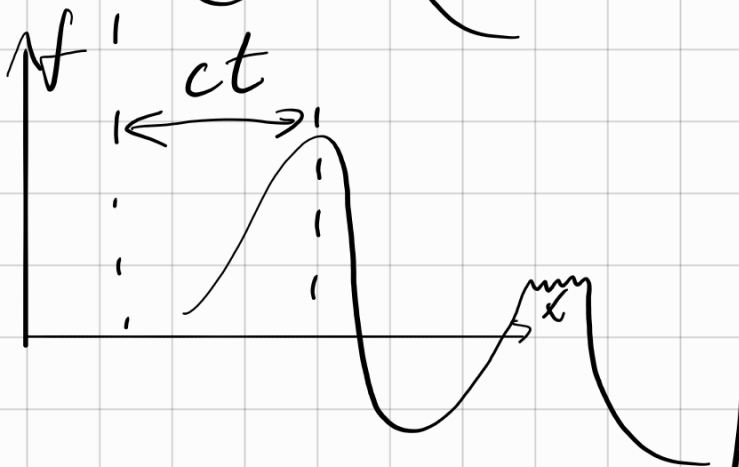
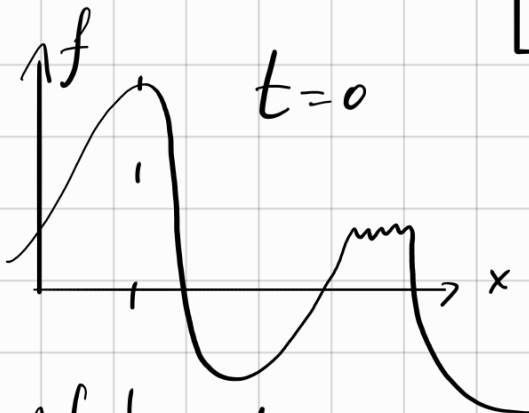
$$\int m \frac{\partial^2 \xi}{\partial t^2} = \frac{\partial F}{\partial x} dx$$

$$\int m = \rho S dx$$

$$\frac{F(x)}{S} = \sigma(x) = E \epsilon(x) = E \frac{\xi(x+dx, t) - \xi(x, t)}{dx} = E \frac{\partial \xi}{\partial x}$$

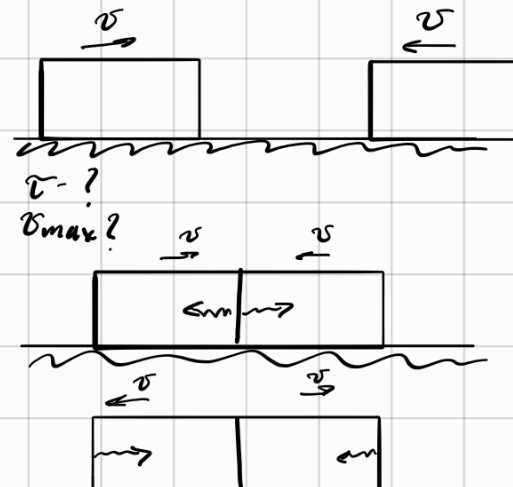
$$\rho S dx \frac{\partial^2 \xi}{\partial t^2} = S \cdot E \frac{\partial^2 \xi}{\partial x^2} dx$$

$$\boxed{\frac{\partial^2 \xi}{\partial t^2} = c^2 \frac{\partial^2 \xi}{\partial x^2}} \quad \text{волновое уравнение } c = \sqrt{\frac{E}{\rho}}$$

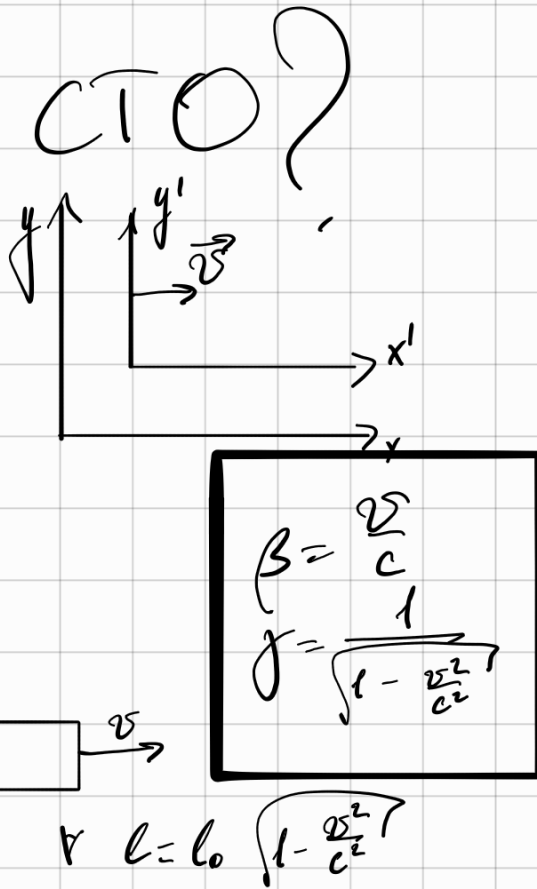


$$f(x - ct)$$

13.39



$$\tau = \frac{2l}{c} = 2l \sqrt{\frac{\rho}{E}} \approx 4 \cdot 10^{-5}$$



$$E \frac{\Delta l}{l_0} = T_{\text{уп}}$$

$$\Delta l = v_{\text{max}} \frac{T}{2}$$

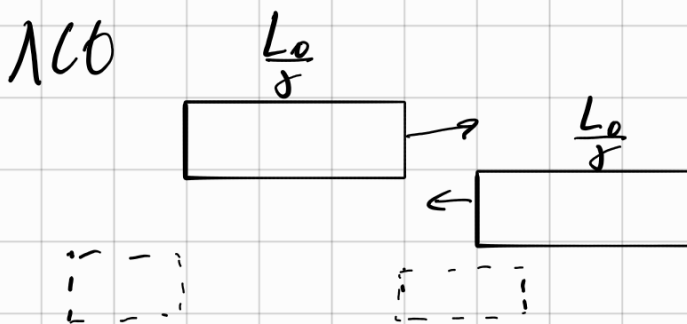
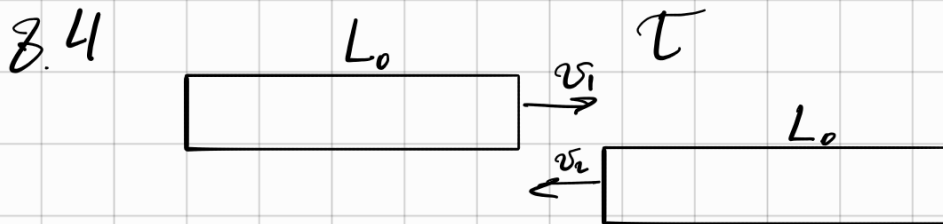
$$\frac{v_{\text{max}}}{c} = \frac{T_{\text{уп}}}{E}$$

$$v_{\text{max}} \approx 5 \text{ м/с}$$

$$x' = \frac{x - vt}{\sqrt{1 - \frac{v^2}{c^2}}} \quad y' = y \quad t' = \frac{t - \frac{v}{c^2} x}{\sqrt{1 - \frac{v^2}{c^2}}}$$

⇓

- 1) Задержка времени
- 2) Сокращение длины
- 3) Относительность одновременности



СО линейки



$$\tau v = L_0 - L_0/\gamma$$

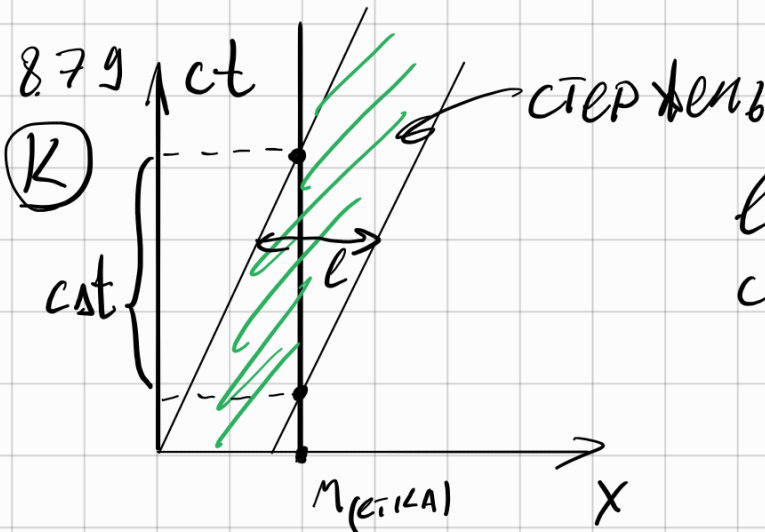
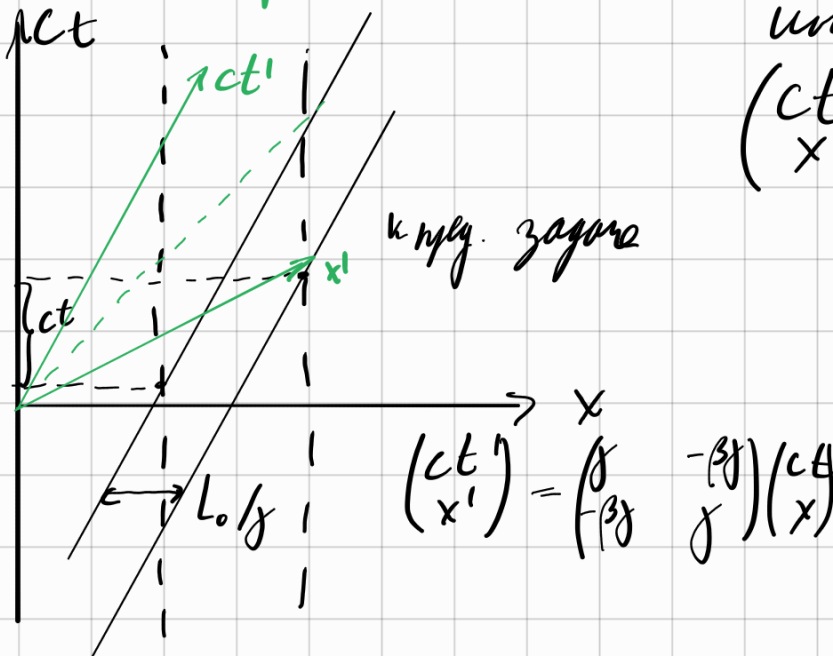
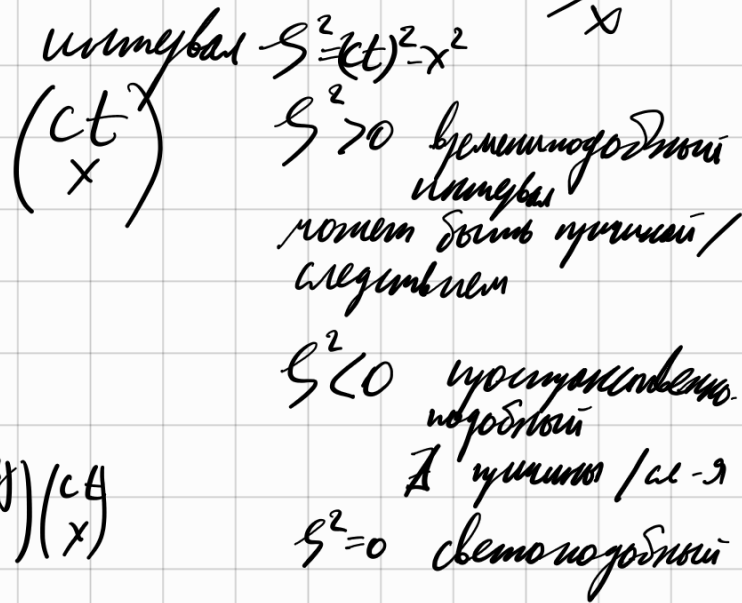
$$v \tau = L_0 - L_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$v = \frac{2L_0\tau}{\tau^2 + \frac{L_0^2}{c^2}}$$

no Inom
ridbyblad
Bleasat na kanskygumpe?
O k mo, Fismen?
Fismen???

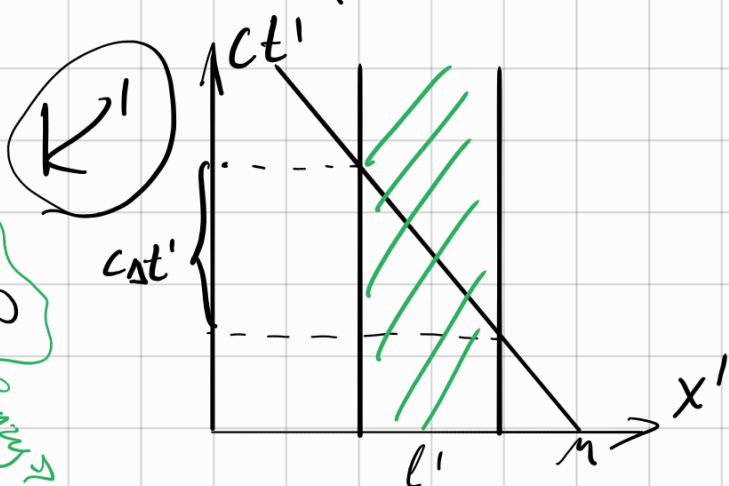


из АВВ логика
из АВС пизда



$$l = v_{\Delta t}$$

СОКРАЩ. ДЛИНА



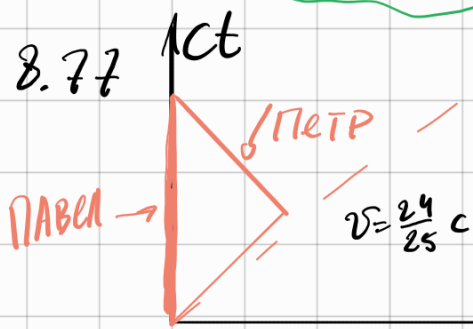
$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{\Delta t'}{\Delta t}$$



5

$$\frac{v}{c} = \sqrt{1 - \frac{1}{\gamma^2}} = \sqrt{1 - \left(\frac{\Delta t}{\Delta t'}\right)^2}$$

$$l' = c \sqrt{\Delta t'^2 - \Delta t^2} = c \cdot 4 \cdot 10^{-9} \approx 1.2 \text{ н.м.}$$

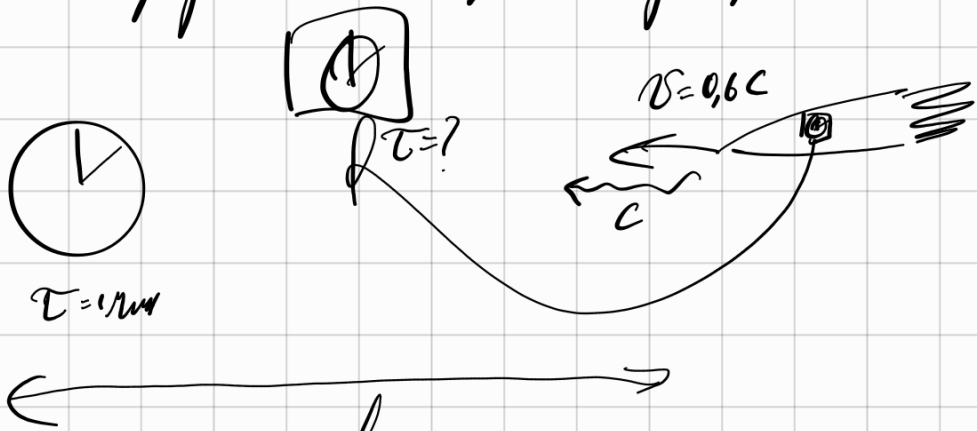


$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{25}{7}$$

$$\Delta t = \gamma \Delta t' = \frac{25}{7} \cdot 14 \text{ лет} = 50 \text{ лет}$$

Получил Близнецов, где пред: СО ПЕТРА и неру.

8.29



t_0 - начало опыта.

t'_0 - начало сист. для опыта

t_1 - конец опыта.

t'_1 - конец сист. для опыта.

$$t_1 - t_0 = \tau$$

$$t_0 - t'_0 = \frac{l}{c}$$

$$t_1 - t'_1 = \frac{l - (t'_1 - t'_0) v}{c} = t_0 - t'_0 - \frac{v}{c} (t'_1 - t'_0)$$

$$\tau = t_1 - t_0 = (t'_1 - t'_0) \left(1 - \frac{v}{c}\right)$$

КЛАССИЧЕСКИЙ ЭФ. ДОПЛЕРА

$t'_1 - t'_0 = \frac{\tau}{1 - \frac{v}{c}}$ - промежуток между сигналами излученным с ракетой по часам Земли

$$\tau' = \frac{t'_1 - t'_0}{\gamma} = \frac{\tau}{1 - \frac{v}{c}} \cdot \sqrt{1 - \frac{v^2}{c^2}} = \tau \sqrt{\frac{1 + \frac{v}{c}}{1 - \frac{v}{c}}} \quad \text{— релятивистский эф. Доплера}$$

$$v_x = \frac{dx}{dt} \quad v'_x = \frac{dx'}{dt'}$$

$$v_y = \frac{dy}{dt} \quad v'_y = \frac{dy'}{dt'} = \frac{dy}{dt}$$

$$V = \frac{v + u}{1 + \frac{vu}{c^2}} \quad \text{релятивистский закон сложения скоростей}$$

$$\vec{p} = m \vec{v}$$

$$p = \gamma m v$$

$$u - \text{импульс} \begin{pmatrix} E/c \\ \vec{p} \end{pmatrix}$$

$$\vec{F} = \frac{d\vec{p}}{dt}$$

$$\vec{F} = \frac{d\vec{p}}{dt}$$

$$E = \gamma mc^2 = E_0 + K$$

при $v=0 \quad E_0 = mc^2$

8.59 $E = 10 \Gamma \Theta B$

$$p^+ \rightarrow \quad E' - ? \quad \leftarrow p^+$$

$$\Gamma \Theta B = 1,6 \cdot 10^{-19} \text{ Па}$$

$$E_0 = m_p c^2 \approx \Gamma \Theta B$$

$$\left(\frac{E}{c}\right)' = \gamma \left(\frac{E}{c} - \beta p\right)$$

$$E' = \gamma E - \gamma \beta p c$$

$$\gamma = \frac{E}{m_p c^2} \approx 10$$

$$\beta = \sqrt{1 - \frac{1}{\gamma^2}} \approx 0,995$$

$$p c = - \gamma m v c = - \gamma m \beta c^2$$

$$E' = \frac{E}{m_p c^2} \left(E + \sqrt{1 - \frac{m_p^2 c^4}{E^2}} \frac{E}{m_p c^2} \sqrt{1 - \frac{m_p^2 c^4}{E^2}} m_p c^2 \right) =$$

$$= \frac{E^2}{m_p c^2} \left(1 + 1 - \frac{m_p^2 c^4}{E^2} \right) = \frac{E^2}{m_p c^2} \left(2 - \frac{m_p^2 c^4}{E^2} \right) \approx 212 \Gamma \Theta B$$

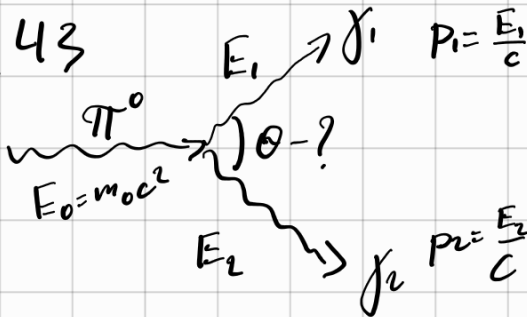
$$s^2 = (ct)^2 - x^2$$

$$I^2 = \left(\frac{E}{c}\right)^2 - p^2 = \frac{m^2 c^4}{c^2 \sqrt{1 - \frac{v^2}{c^2}}} - \frac{m^2 v^2}{1 - \frac{v^2}{c^2}} =$$

$$= \frac{m^2 (c^2 - v^2)}{1 - \frac{v^2}{c^2}} = m^2 c^2$$

$$E^2 - p^2 c^2 = m^2 c^4$$

8.43



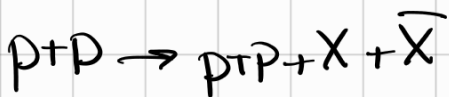
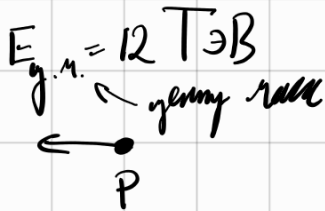
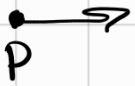
$$E = h\nu$$

$$(m_0 c^2)^2 = (E_1 + E_2)^2 - (\vec{p}_1 + \vec{p}_2)^2 c^2 = E_1^2 + E_2^2 + 2E_1 E_2 - c^2 (p_1^2 + p_2^2 + 2p_1 p_2 \cos \theta)$$

$$= 2 E_1 E_2 (1 - \cos \theta)$$

$$\cos \theta = 1 - \frac{m_0^2 c^4}{2 E_1 E_2}$$

T 15.1



m_X - ?

1) bumperное столкновение

$$E_1 = E_2 = 6 T \Delta B$$

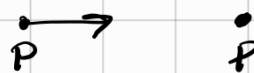
$$p = 0$$

$$E_{\text{г.л.}} = (2 m_p + 2 m_X) c^2$$

$$m_X = \frac{E_{\text{г.л.}}}{2 c^2} - m_p \approx 6 T \Delta B / c^2$$

2) протон. мимено

$$E = 6 T \Delta B = \frac{E_{\text{г.л.}}}{2}$$



$$M = 2(m_p + m_X)$$

$$M^2 c^4 = \left(\frac{E_{\text{г.л.}}}{2} + m_p c^2\right)^2 - p^2 c^2 \Rightarrow$$

Лаз мимено
б мимено корноу

Science! Computer



инвариант энергии импульса

$$\left(\frac{E_{y.1}}{c}\right)^2 - p^2 = m_p^2 c^2$$

$$(p_0^2 - \left(\frac{E_{y.1}}{c}\right)^2) = m_p^2 c^2$$

$$\Rightarrow \frac{E_{y.1}^2}{4} + E_{y.1} \cdot m_p \cdot c^2 + m_p^2 c^4 - \frac{E_{y.1}^2}{4} + m_p^2 c^4 =$$

$$= E_{y.1} \cdot m_p c^2 + 2 m_p^2 c^4$$

$$4(m_p + m_x)^2 c^4 = E_{y.1} m_p c^2 + 2 m_p^2 c^4 =$$

$$= m_p^2 c^4 \left(\frac{E_{y.1}}{m_p c^2} + 2 \right)$$

$$m_x = \frac{m_p}{2} \left(\sqrt{\frac{E_{y.1}}{m_p c^2} + 2} - 2 \right)$$

$$\frac{d\vec{p}}{dt} = \vec{F} = \frac{d}{dt} (\gamma m \vec{v}) = m \left(\vec{v} \frac{d\gamma}{dt} + \gamma \frac{d\vec{v}}{dt} \right) =$$

$$= \gamma m \vec{a} + m \vec{v} \frac{1 - \frac{v^2}{c^2}}{-2 \left(1 - \frac{v^2}{c^2} \right)^3} \frac{d\vec{v}}{dt} = \gamma m \vec{a} + \gamma^3 m \frac{\vec{v}}{c^2} (\vec{v} \cdot \vec{a})$$

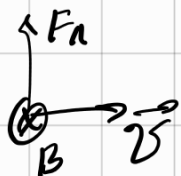
$$(\vec{F} \cdot \vec{v}) = \gamma m (\vec{a} \cdot \vec{v}) + \gamma^3 m \frac{v^2}{c^2} (\vec{v} \cdot \vec{a}) = \gamma m (\vec{a} \cdot \vec{v}) \left(1 + \gamma^2 \frac{v^2}{c^2} \right) =$$

$$= \gamma m (\vec{a} \cdot \vec{v}) \left(1 + \frac{\frac{v^2}{c^2}}{1 - \frac{v^2}{c^2}} \right) = \gamma^3 m (\vec{a} \cdot \vec{v}) ; (\vec{a}; \vec{v}) = \frac{(\vec{F}; \vec{v})}{\gamma^3 m}$$

$$\boxed{a = \frac{1}{\gamma m} \left(\vec{F} - \vec{v} \frac{(\vec{F} \cdot \vec{v})}{c^2} \right)} \quad \vec{a} \perp \vec{F}$$

$$1) \vec{F} \perp \vec{v} \quad (\vec{F} \cdot \vec{v}) = 0 \quad \vec{a} = \frac{\vec{F}}{\gamma m}$$

$$\vec{F}_n = q [\vec{v} \times \vec{B}]$$

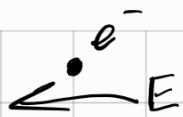


$$a = \frac{v^2}{R} = \frac{q v B}{\gamma m}$$

$$R = \frac{\gamma m v}{q B}$$

$$2) \vec{F} \parallel \vec{v} \quad \vec{a} = \frac{\vec{F}}{\gamma m} \left(1 - \frac{v^2}{c^2} \right) = \frac{\vec{F}}{\gamma^3 m}$$

T 15.2



$$\vec{F} = e\vec{E} = \text{const}$$

$$l = 1 \text{ м.} \quad \tau = 5,8 \text{ нс.}$$

$$\frac{K}{m_e c^2} = ?$$

$$P = Fv = \gamma m v^2 = mc \sqrt{\gamma^2 - 1}$$

$$A = Fl = K$$